MROD-YOLO: Multimodal Joint Representation for Small Object Detection in Remote Sensing Imagery via Multiscale Iterative Aggregation

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***Abstract*— Accurate and efficient small object detection using multimodal remote sensing images on resource-constrained aerial platforms is a challenging task. Most existing solutions rely on complex networks to extract object features, which often incurs significant computational overhead. In this article, we propose MROD-YOLO, a novel multimodal object detection framework designed to enhance the accuracy and efficiency of remote sensing imagery analysis. First, we design a multimodal joint representation network (MJRNet), which employs a global context attention block (GCB) to maximize feature retention during fusion. MJRNet integrates complementary features from multimodal images, addressing the limitations of single-modal methods in complex scenarios. Furthermore, we introduced the receptive field expansion mechanism (RFEM) to the spatial feature representation of backbone networks. Finally, a multiscale iterative aggregation (MSIA) module is developed to refine feature interactions and improve the detection of small objects in challenging environments. By replacing the path aggregation network (PANet) with a feature pyramid network (FPN), the model ensures effective feature preservation for texture details. Experimental results show that MROD-YOLO achieves an accuracy of 77.9% mAP50, which is 4.2% higher than the SOTA large models CFT. MROD-YOLO provides a promising solution for real-time, high-precision object detection in multimodal remote sensing applications.**

***Index Terms*— Feature aggregation, multimodal joint representation, object detection, remote sensing imagery.**

I. Introduction

With the advancement of remote sensing technology and the increasing availability of spatial data, remote sensing object detection has gradually become a key research direction in remote sensing image analysis [1], [2], [3]. Remote sensing object detection aims to automatically identify and localize specific targets, such as vehicles, ships, and buildings, within spatial imagery [4], [5]. This capability addresses practical requirements across various fields, including military surveillance, disaster assessment, urban planning, and environmental monitoring. Remote sensing images are characterized by a wide field of view, high resolution, and extensive coverage. However, they also pose significant challenges due to complex background features, small object scales, and a wide variety of target types.

Currently, remote sensing object detection primarily focuses on optical images. Although optical remote sensing imaging can capture detailed target information, it is highly susceptible to environmental factors. Many researchers have achieved promising detection accuracy in optical remote sensing object detection [6], [7], [8], [9]. However, in more challenging scenarios, such as nighttime, rainy, or foggy conditions, relying solely on optical remote sensing images for object detection often fails to achieve accurate target identification. Infrared remote sensing technology can acquire imagery in low-light conditions, such as nighttime and overcast environments. Furthermore, it possesses the capability to penetrate clouds and smoke, enabling all-weather object detection [10], [11], [12]. Infrared remote sensing images are primarily based on the temperature distribution of objects, lacking the color and texture features present in visible light images. This limitation makes it challenging to distinguish objects with similar temperatures but different properties, leading to a higher likelihood of false detections or missed detections. Additionally, infrared remote sensing images typically have lower spatial resolution, making it difficult to discern fine details of ground features.

Most object detection methods are specifically designed and tailored for single-modal data. Given the complementary information that exists across different modalities, single-modal detection methods still exhibit significant limitations. In contrast, multimodal remote sensing data can provide more comprehensive information. How to effectively acquire high-quality multimodal fusion information and comprehensively understand the characteristics of target objects to enhance object detection capabilities remains a major challenge in fusion detection technology.

In recent years, to meet the demand for detection accuracy, deep learning-based methods have achieved significant progress in the field of object detection, leveraging powerful computational capabilities and deep network

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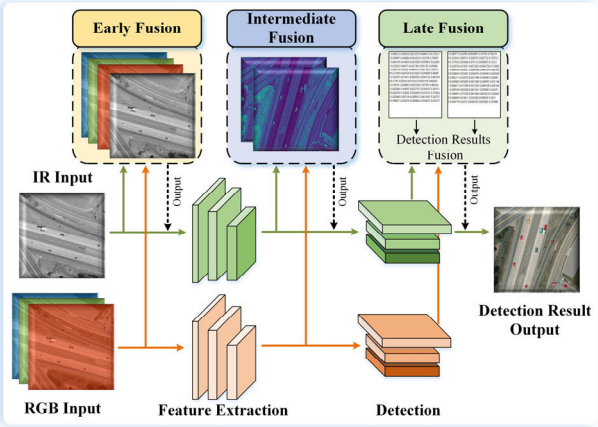


Fig. 1. Diagram of different fusion strategies.

architectures [13], [14], [15]. To enhance the detection performance of deep learning models on multimodal data, a common approach is to integrate the acquired multimodal data at different levels. Currently, there are three fusion paradigms: early fusion, intermediate fusion, and late fusion, as illustrated in Fig. 1 [16], [17], [18], [19].

The diagram illustrates the placement of fusion operations within the detection network under different fusion strategies. For remote sensing small object detection, selecting an appropriate multimodal fusion detection method requires considering the characteristics of small objects and the complementarity of multimodal data. The early fusion strategy directly fuses images from different modalities at the input stage, using the fused data as a unified input for subsequent network processing. This approach effectively preserves the original information between modalities and has relatively low computational overhead.

In this work, we propose MROD-YOLO, an efficient real-time object detection framework for multimodal remote sensing images, aimed at fully utilizing the complementary characteristics of multimodal data to achieve accurate detection and rapid response of targets. Throughout this process, we place particular emphasis on the computational complexity of the algorithm. By designing a lightweight model and optimizing the fusion strategy, we aim to significantly improve detection accuracy without introducing additional computational overhead. At the same time, we maintain efficient inference speeds to meet the stringent requirements of real-time processing tasks.

To make full use of the complementarity between different modalities, we propose a multimodal joint representation network (MJRNet) that enhances the ability of the network to learn detailed features, thereby improving the detection accuracy of multimodal remote sensing images. By analyzing and evaluating different fusion strategies, we selected the early fusion approach, which is more conducive to small object detection. This method better preserves multimodal feature details, reducing information loss and feature deviation. To enhance the feature representation of small objects, we integrated a receptive field expansion mechanism (RFEM) into the backbone network to strengthen its feature extraction capability. Additionally, we developed a multiscale iterative aggregation (MSIA) module to facilitate the efficient fusion of high-level and low-level features. By incorporating a channel attention mechanism to emphasize local details, preventing the fine-grained features of small objects from being diluted during the fusion process. Finally, for small object detection, maintaining high spatial resolution and sufficient semantic strength is crucial. Therefore, we replaced the path aggregation network (PANet) with a feature pyramid network (FPN) to better meet these requirements. FPN utilizes a relatively simple and straightforward hierarchical fusion mechanism to ensure sufficient semantic features at shallow layers. This approach prevents small object features from becoming less prominent during the fusion process due to overly complex multipath information flows.

In summary, the main contributions of this study are as follows.

1) We propose an efficient multimodal fusion detection framework, MROD-YOLO, which combines early fusion with an MSIA module to dynamically and accurately optimize the interaction of multimodal features. This MROD-YOLO significantly improves detection accuracy while effectively reducing computational overhead.
2) Considering the need for high-quality detection and low computational cost, we introduce an efficient MJRNet. The MJRNet employs a global context attention block (GCB) to dynamically weight information from different modalities, maximizing the preservation of original multimodal details.
3) We introduce the MSIA module into the multimodal object detection framework for the first time. Our method employs a multiscale channel attention (MCA) mechanism to dynamically adjust the interactions between high-level and low-level features. By optimizing feature quality through iterative attention fusion, it enhances the model's ability to perceive small objects and targets in complex backgrounds.

The remainder of this article is organized as follows. Section II offers a succinct overview of the relevant literature. The main content of Section III includes a detailed description of the proposed method. The experimental results are presented in Section IV. Section V presents the conclusion of the article.

II. Related Works

In this section, we provide a brief overview of pertinent methods in single-modal object detection and multimodal object detection.

A. Single-Modal Object Detection

With the rapid development of deep neural networks, deep learning-based object detection technology has achieved remarkable progress and has been widely applied in intelligent unmanned systems, such as drones and autonomous vehicles. Currently, deep learning-based object detection methods can be broadly categorized into one-stage and two-stage detectors. Two-stage detectors, represented by R-CNN [20], employ a region proposal network (RPN) to generate candidate regions,

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which are then classified and refined through bounding box regression. The introduction of Fast R-CNN [21] and Faster R-CNN [13] significantly improved detection efficiency and accuracy. The YOLO series [15], [22], [23], [24], [25] and SSD [26] directly regress bounding boxes and class labels on the image, enabling real-time object detection. Transformer-based detectors, such as DETR [14], have gained attention due to their advantages in global context modeling, further advancing object detection technology.

Remote sensing images typically cover a wide range of surface information, with detected objects exhibiting significant variations in size and scale. To address the above issues, TBNet [27] proposed a texture- and boundary-aware RCNN detection network. The texture-aware module is designed to explore image details, while the boundary fusion module emphasizes spatial positioning. To apply the YOLO algorithm to remote sensing datasets, it is essential to enhance the detection accuracy for small objects. Gong et al. [28] incorporated attention mechanisms and additional detection heads into the traditional YOLOv5 architecture, enabling precise detection of small objects. Lin et al. [29] proposed a more efficient multiscale feature fusion network based on the YOLOv7 architecture to enhance the network's multiscale modeling capability. Zhang et al. [30] designed an efficient feature extraction module based on channel attention and average pooling to address multiscale object detection. To address the challenges of cluttered target distribution and small object features in SAR remote sensing image detection, Liu et al. [31] introduced the bi-former attention mechanism to improve small object detection and utilized the KLD loss function to enhance accuracy.

Although these detectors have demonstrated excellent performance, they rely solely on single-modality data for remote sensing object detection. To overcome the limitations of optical images in rainy and foggy conditions, infrared remote sensing images, with their all-weather detection capability, have been widely utilized. However, the aforementioned methods are limited to a single modality. In contrast, leveraging multimodal data (e.g., infrared and optical images) enables complementary information integration, thereby enhancing target visibility and resolution in complex scenarios. This study selects the YOLO series models as baseline detection models and constructs a multimodal fusion detection framework to improve detection performance in multisource input scenarios.

B. Multimodal Object Detection

Currently, with advancements in sensor technology and the diversification of data acquisition methods, multimodal data has been widely applied in fields such as autonomous driving [32], remote sensing classification, and semantic segmentation [33], [34]. By integrating information from different data sources and fully exploiting the relationships between multimodal data, the limitations of a single data modality can be mitigated. Multimodal object detection employs diverse modality data, meticulously uncovering the complementary characteristics among modalities, thereby augmenting the precision, resilience, and versatility of target detection. In the field of remote sensing detection, various modalities of image data exist (e.g., optical, infrared, SAR, LiDAR). The fusion of modalities with complementary characteristics plays a crucial role in enhancing the performance of various remote sensing tasks [33], [35]. Infrared modality data can detect thermal radiation, and its fusion with optical data can improve detection capabilities in occluded environments, such as those with cloud cover or vegetation. Zhang et al. [36] proposed a fast and accurate object detection method that employs early fusion of multimodal data to reduce computational costs. Research shows that although multimodal fusion can effectively improve detection performance, its detection accuracy and computational efficiency remain insufficient, making it challenging to meet the high-performance requirements of real-time detection tasks.

The features at various stages of an object detection network encompass distinct levels of semantic information and visual detail. As such, feature fusion can be seamlessly integrated at any phase of the network. Even within a single stage, employing different fusion strategies can yield diverse performance outcomes. An architecture capable of harmoniously blending both low-level intricacies and high-level semantic features is essential for achieving superior detection performance. Existing multimodal fusion strategies can be classified based on the fusion approach, with common strategies including early fusion, intermediate fusion, and late fusion [37].

Intermediate fusion typically refers to extracting features separately from raw data of different modalities and then performing cross-modal interaction and fusion at intermediate feature levels. Qingyun et al. [38] proposed a dual-stream multimodal detection network that employs a cross-modal fusion Transformer module to integrate intra-modal and inter-modal features. Dong et al. [46] replaced the Transformer module from [38] with the Mamba module, thereby enhancing the representational consistency of the fused features. Chen et al. [44] employed a dual enhancement mechanism to minimize mutual interference between the two modalities, achieving commendable detection performance in complex low-light environments. Intermediate fusion dynamically integrates multimodal information during the latter stages of feature extraction, leveraging an adaptive weighting mechanism to balance the contributions of each modality, thereby enhancing robustness against noise and modality disparities. However, this strategy inevitably results in the loss of certain high-level semantic information and restricts the flexibility to adapt fusion strategies, ultimately limiting its overall adaptability.

Late fusion achieves a significant enhancement in detection accuracy by integrating the predictive outcomes generated from distinct modality detection networks. Chen et al. [32] introduced a late-stage fusion technique, ProbEn, based on a probabilistic ensemble. Through probabilistic marginalization, ProbEn adeptly addresses the issue of detection omissions. However, the late fusion detection strategy, which integrates modalities only at the final stage of detection, fails to fully leverage the complementary nature of multimodal features during extraction and inference, thereby limiting the efficacy of fusion. Furthermore, the process of merging detection results in late fusion consumes substantial computational resources.

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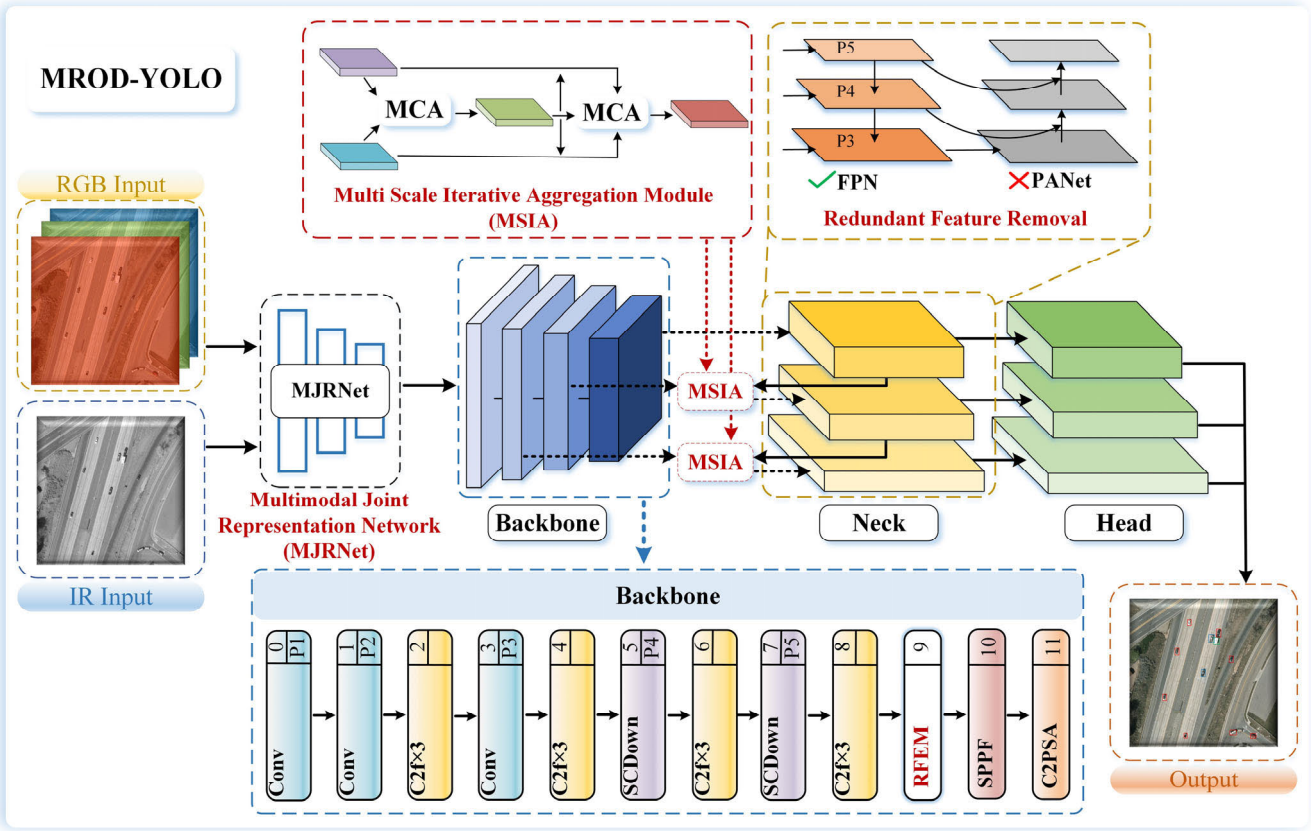


Fig. 2. Overall network framework, MROD-YOLO. Our new contributions include: 1) MJRNet combines RGB and IR images to maximize complementary information acquisition; 2) RFEM and C2PSA expand the receptive field of backbone networks while enhancing feature representation and contextual information; 3) MSIA iteratively aggregates the outputs of the backbone network and neck network to efficiently fuse and enhance the raw semantic and spatial information through the neck network; and 4) replace PANet with FPN and remove redundant features.

In contrast, early fusion integrates multimodal images after feature extraction, combining information at the lower feature level. This approach reduces redundant computations while mitigating potential information loss during subsequent feature extraction, thereby enhancing the detection of small objects. In this article, we propose an early fusion detection network, MROD-YOLO. First, we construct a multimodal joint network based on global contextual attention, effectively fusing complementary information from different modalities while preserving feature details and minimizing information loss. Subsequently, an MSIA module is designed to efficiently integrate features across various dimensions. Finally, a redundant feature elimination strategy is employed to retain semantic information from shallow layers. Moreover, our method requires only a single feature extraction step, enhancing accuracy while significantly reducing computational costs.

III. Method

The comprehensive framework of our proposed MROD-YOLO network, designed for multimodal remote sensing object detection, is illustrated in Fig. 2. First, we explored different image fusion methods and proposed an MJRNet. By performing early fusion of infrared and visible-light modality images at the front end of the backbone network, we fully utilized multimodal data to achieve complementary feature integration.

Second, we introduced the RFEM and cross-stage partial spatial attention (C2PSA) into the backbone network. These mechanisms dynamically adjust the receptive field range of the network while enhancing spatial feature representation, significantly improving the detection capability for small objects. Subsequently, we added a multiscale information iterative aggregation module at the connection between the backbone network and the neck. This module aggregates contextual information from different receptive fields for targets of varying scales, thereby enhancing the quality of feature fusion. Finally, we removed the PANet structure [39] from the feature fusion network and replaced it with an FPN network [40], emphasizing the supplementation of semantic information to detailed features. This enhancement makes low-level features more distinguishable, thereby preventing accuracy degradation.

A. Multimodal Joint Representation Network

The richer the modality data used for information complementarity, the higher the accuracy of object detection, enabling the detection model to achieve a more comprehensive perception. MJRNet effectively leverages information provided by different sensors and data sources, thereby enhancing the performance of detection tasks. For remote sensing object detection tasks, intermediate fusion can lead to information

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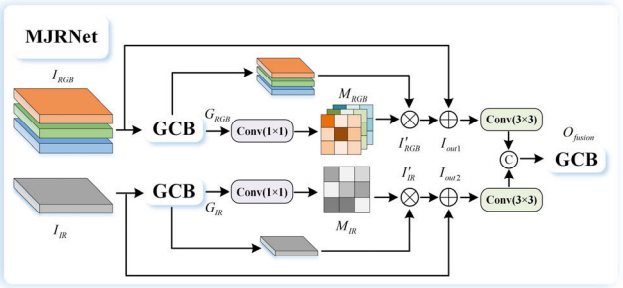


Fig. 3. Schematic of MJRNet.

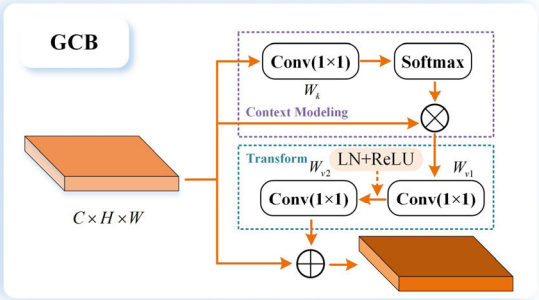


Fig. 4. Architecture of the GCB. $C \times H \times W$ denotes a feature map with channel number C, height H, and width W.

loss or distortion during feature extraction. For small objects in remote sensing, any information loss can severely impact detection performance. Similarly, late fusion only merges the detection results at the output stage, neglecting the complementary low-level features between different modalities, while also increasing computational time.

We propose an MJRNet to enhance the interaction between different modalities, which improves our method's ability to capture detailed features, thereby increasing the detection accuracy for small objects. Unlike traditional early fusion methods such as simple concatenation or weighted fusion, we employ a GCB to automatically focus on critical information regions and dynamically weight information from different modalities, achieving high-accuracy multimodal fusion detection.

As shown in Fig. 3, the input modalities of RGB and infrared are represented as $I_{\mathrm{RGB}}, I_{\mathrm{IR}} \in \mathbb{R}^{C\times H\times W}$, respectively. They are fed into the GCB module to capture long-range dependencies at different locations, resulting in outputs G_{RGB} and G_{IR}

$$G_{\mathrm{RGB}} = \mathrm{GCB}(I_{\mathrm{RGB}}), G_{\mathrm{IR}} = \mathrm{GCB}(I_{\mathrm{IR}}). \tag{1}$$

Then, a mask generation process is employed to extract and enhance distinctive feature regions within the image, significantly highlighting critical spatial areas between modalities. This process reveals the spatial correspondence of information across different modalities. The mask generation is defined as follows:

$$M_{\mathrm{RGB}} = \mathrm{Conv1}(I_{\mathrm{RGB}}), M_{\mathrm{IR}} = \mathrm{Conv2}(I_{\mathrm{IR}}) \tag{2}$$

where Conv1 and Conv2 represent the 1×1 convolution operations applied to the RGB and infrared modalities, respectively.

The generated masks are used to weight the different modality images. Specifically, element-wise multiplication is performed to weight the corresponding locations of the mask and the image. After weighting, deeper spatial information between different modality images is produced by

$$I'_{\mathrm{RGB}} = G_{\mathrm{RGB}} \otimes M_{\mathrm{RGB}}, I'_{\mathrm{IR}} = G_{\mathrm{IR}} \otimes M_{\mathrm{IR}} \tag{3}$$

where $\otimes$ refers to the element-wise matrix multiplication.

Using residual connections, the weighted features are concatenated with the original features, followed by convolution processing to obtain the final features for each modality. The process is defined as follows:

$$I_{\mathrm{out1}} = \mathrm{Conv3}\left(I_{\mathrm{RGB}} + I'_{\mathrm{RGB}}\right), I_{\mathrm{out2}} = \mathrm{Conv4}\left(I_{\mathrm{IR}} + I'_{\mathrm{IR}}\right) \tag{4}$$

where Conv3 and Conv4 denote the 3×3 convolution operations.

Finally, the feature maps of the RGB and infrared modalities are concatenated along the channel dimension, and the GCB module is applied to further adjust the weights between channels, achieving feature fusion. The fused feature representation is defined as follows:

$$O_{\mathrm{fusion}} = \mathrm{GCB}(I_{\mathrm{out1}} \copyright I_{\mathrm{out2}}) \tag{5}$$

where © represents concatenation along the channel dimension. The fusion result is then fed into the backbone network, generating multidimensional features.

The GCB module, as shown in Fig. 4, consists of three parts.

1) Global attention pooling is utilized for context modeling. A 1×1 convolution W_k and the Softmax function is applied to obtain attention weights, followed by pooling to extract global context features.
2) To reduce the number of module parameters, a bottleneck transform module is used to replace traditional 1×1 convolution. Additionally, layer normalization is incorporated within the bottleneck transform (before ReLU) to act as a regularizer.
3) Broadcast element-wise addition is used to aggregate global contextual features into the features at each location.

The unique design of the GCB module lies in its integration of the strengths of non-local networks (NLNets) and squeeze-and-excitation networks (SENets), while achieving efficient and lightweight global context modeling through innovative structures and mechanisms. Traditional NLNet generates attention maps by computing pairwise relationships between each query position and all other positions, which is computationally intensive and redundant. In contrast, the GCB module directly generates a global attention map that is independent of query positions via global attention pooling, thereby significantly reducing computational complexity. Additionally, the introduction of a two-layer bottleneck structure not only reduces the number of parameters but also enhances the model's optimization capability through layer normalization, enabling more effective capture of complex inter-channel dependencies.

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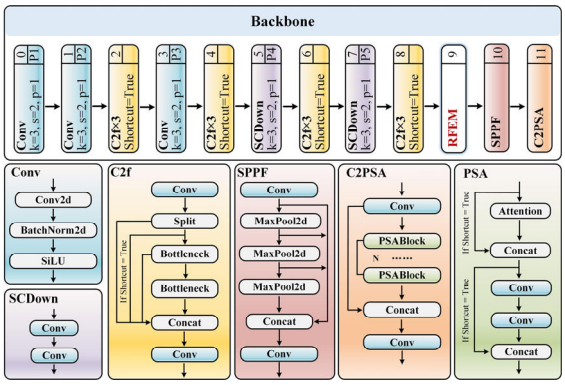


Fig. 5. Backbone structure of MROD-YOLO.

The detailed architecture of the GCB module can be formulated as follows:

$$y_i = x_i \oplus \left(W_{v2}\mathrm{ReLU}\left(\mathrm{LN}\left(W_{v1} \sum_{j=1}^{N_p} \frac{e^{W_k x_j}}{\sum_{m=1}^{N_p} e^{W_k x_m}} \otimes x_j \right) \right) \right) \tag{6}$$

where x_i and y_i represent the input and output of the module, respectively. $i = 1, 2 \ldots N_p$ denotes the index of the query position, N_p is the number of positions in the feature map, and j enumerates all possible positions. $W_{v2}\mathrm{ReLU}(\mathrm{LN}(W_{v1}(\cdot)))$ is the bottleneck transform module. W_{v1} and W_{v1} represent the 1×1 convolution operations. $\oplus$ represents the broadcast element-wise addition.

B. Backbone Network Improvement

As shown in Fig. 2, MJRNet feeds the fused result into the backbone network, which extracts shallow texture features and deep semantic information from the image. Our improvements to the backbone network are illustrated in Fig. 5. A critical challenge in extracting features for small objects in remote sensing images lies in the complexity of the background and the significant differences in background characteristics between different targets.

To dynamically adjust the receptive field and more effectively address the differences in background information required by various targets, we embedded an RFEM before the SPPF layer. Similarly, to enhance feature representation and enable the network to better focus on useful regions in the image, we introduced the C2PSA module. Through the C2PSA mechanism, feature maps are automatically weighted to enhance spatial feature representation and contextual information. This aids the model in better recognizing and localizing small objects.

Our RFEM is illustrated in Fig. 6. We obtain background information features from different regions of the input feature X through a series of decoupled deep convolutions with varying receptive fields, as expressed by the following equation:

$$U_0 = X, U_{i+1} = \mathcal{F}_i^{\mathrm{dw}}(U_i) \tag{7}$$

where $\mathcal{F}_i^{\mathrm{dw}}(\cdot)$ represents a depth-wise convolution operation with a kernel size of k_i and a dilation rate of d_i.

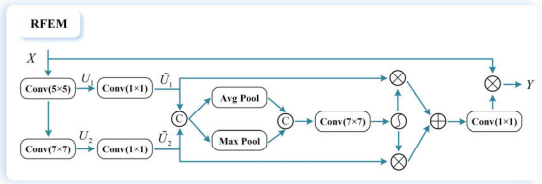


Fig. 6. RFEM structure of MROD-YOLO.

Assuming the number of decoupled convolution kernels is N, the depth-wise convolution is followed by a point-wise convolution (PWConv) to capture inter-channel information while simultaneously achieving channel dimensionality reduction:

$$\tilde{U}_i = \mathcal{F}_i^{1\times1}(U_i), i \in [1, N]. \tag{8}$$

By applying large convolutional kernels with different scales to the input features, spatial selection is performed to focus on the key background information of the target in the spatial domain. First, the features obtained from (8) are concatenated to form $\tilde{U} = \mathrm{concat}(\tilde{U}_1, \ldots, \tilde{U}_i)$. Subsequently, channel-wise average pooling $P_{\mathrm{avg}}(\cdot)$ and max pooling $P_{\max}(\cdot)$ are applied to extract spatial relationships. To enable the interaction of information from different spatial descriptions, the pooled features are concatenated and transformed into the desired spatial attention feature map S through a large convolutional kernel $\mathcal{F}_i^{7\times7}(\cdot)$, which can be expressed as

$$S = \mathcal{F}_i^{7\times7}\left(P_{\mathrm{avg}}(\tilde{U}) \copyright P_{\max}(\tilde{U})\right). \tag{9}$$

Each spatial attention feature map S_i is passed through a Sigmoid activation function to obtain the independent spatial selection attention coefficients $\tilde{S}_i$ corresponding to the decoupled convolution kernels. Subsequently, the sequence of decoupled convolution kernels is weighted with the corresponding attention coefficients and fused through a 1×1 convolution to obtain the final attention feature $SA = \mathcal{F}_i^{1\times1}(\sum_{i=1}^{N}(\tilde{S}_i \cdot \tilde{U}_i))$. Finally, the output feature is obtained by the element-wise matrix multiplication of the input feature Y and the final attention feature SA, expressed as

$$Y = X \otimes SA. \tag{10}$$

C. MSIA Module

After obtaining multilevel feature representations from the backbone network, it is necessary to efficiently fuse and enhance the raw semantic and spatial information through the neck network. However, current research primarily focuses on simple feature-level concatenation between the output of the backbone and the input to the neck. Unfortunately, directly concatenating feature maps along the channel or spatial dimensions introduces substantial redundant information and low-quality feature channels into the fused features, which is detrimental to subsequent detection tasks. It is worth noting that small objects in remote sensing images typically appear in local regions of the image and tend to become weakly represented in deep feature maps due to multiple down-sampling operations. When high-level semantic features are simply concatenated with low-level fine-grained features (which are more sensitive to small objects), a significant

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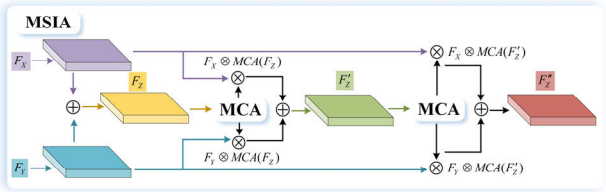


Fig. 7. Multiscale iterative aggregation module.

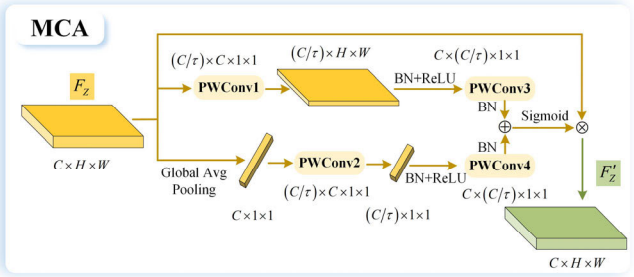


Fig. 8. MCA structure of MSIF.

number of deep feature channels may dominate. As a result, the fine-grained features that are beneficial for small objects can have their signals diluted or overshadowed in the fused feature map.

To address this issue, we introduce the MSIA module, as illustrated in Fig. 7. High-level semantic features and low-level texture features are fed into an MCA module to integrate the input features. The quality of the initial integration significantly impacts the final fusion outcome. Then, the MCA module is iteratively applied, where the integrated features undergo additional attention-based fusion to obtain the final high-quality fused feature representation.

The MCA module we use is illustrated in Fig. 8. To enable the network to simultaneously focus on global information and local details, the module comprises two network branches: a global context extraction branch and a local context extraction branch. The local context is extracted through PWConv, which facilitates inter-channel interaction at each spatial location to capture local information. This process preserves the spatial structure of the input features, emphasizing local details effectively.

The overall computation process of the MSIA module is as follows:

$$\begin{aligned} F_Z'' &= \mathrm{MSIA}(F_X, F_Y) \\ &= \left(F_X \otimes \mathrm{MCA}\left(F_Z'\right)\right) \oplus \left(F_Y \otimes \mathrm{MCA}\left(F_Z'\right)\right) \end{aligned} \tag{11}$$

where F_X and F_Y refer to the low-dimensional feature map and high-dimensional feature map, respectively. F_Z'' is the fused feature representation. F_Z' denotes the integrated input features, which can be expressed as

$$F_Z' = (F_X \otimes \mathrm{MCA}(F_Z)) \oplus (F_Y \otimes \mathrm{MCA}(F_Z)) \tag{12}$$

where $F_Z = F_X \oplus F_Y$.

Assuming the input to the MCA module is F_Z, the output $\tilde{F}_Z$ can be expressed as

$$\tilde{F}_Z = F_Z \otimes \sigma(g(F_Z) \oplus l(F_Z)) \tag{13}$$

where σ is the Sigmoid activation function, $g(F_Z)$ refers to global channel context, $l(F_Z)$ denotes local channel context, which can be expressed as follows:

$$l(F_Z) = \mathcal{B}\left(\mathcal{F}_3^{\mathrm{PW}}\left(\delta\left(\mathcal{B}\left(\mathcal{F}_1^{\mathrm{PW}}(F_Z)\right)\right)\right)\right) \tag{14}$$

$$g(F_Z) = \mathcal{B}\left(\mathcal{F}_4^{\mathrm{PW}}\left(\delta\left(\mathcal{B}\left(\mathcal{F}_2^{\mathrm{PW}}\left(\mathcal{G}_{\mathrm{avg}}(F_Z)\right)\right)\right)\right)\right) \tag{15}$$

where $\mathcal{B}$ represents batch normalization, δ denotes the ReLU activation function, $\mathcal{F}_1^{\mathrm{PW}}$, $\mathcal{F}_3^{\mathrm{PW}}$, $\mathcal{F}_2^{\mathrm{PW}}$, and $\mathcal{F}_4^{\mathrm{PW}}$ are PWConvs used for feature compression and expansion. $\mathcal{G}_{\mathrm{avg}}$ refers to global average pooling.

D. Redundant Feature Removal Strategy

As shown in Fig. 2, the most used fusion network in the NECK module is PANet. Built upon the FPN, PANet introduces a bottom-up feature aggregation path, enabling shallow fine-grained features to further integrate with high-level semantic features during the upward propagation process. However, the additional bottom-up fusion path introduced in PANet increases the complexity of the feature flow. In remote sensing small object detection, excessive intertwining of features may dilute the sensitivity of shallow features to small objects, thereby affecting detection performance. Compared to PANet, the structure of FPN is simpler, and its design ensures that sufficient semantic features are present at shallow layers (where feature maps are larger), which is beneficial for small object detection. Therefore, we remove the redundant features in PANet and replace them with the more straightforward and effective FPN structure.

IV. Experimental Results

A. Dataset

We conducted experimental validation of our method using the publicly available multimodal remote sensing image datasets, VEDAI [41] and DroneVehicle [42].

1) VEDAI: The VEDAI dataset comprises 1246 high-resolution RGB and infrared images, containing 3640 objects categorized into eight common vehicle classes. Each image, with a resolution of 1024 × 1024 pixels, spans diverse terrains and environments. Notably, vehicles occupy only a small portion of the image pixels, making small object detection particularly challenging.

2) DroneVehicle: The DroneVehicle dataset is a large-scale RGB-IR (visible-light and infrared) multimodal dataset designed for vehicle detection in drone imagery. It encompasses a variety of shooting conditions and scenarios, covering different perspectives, altitudes, and lighting conditions. The dataset includes five vehicle categories: car, truck, bus, van, and freight-car.

B. Evaluation Metrics

We use precision, recall, and mean average precision (mAP) as an evaluation metric to measure the performance of the detector as defined below

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{16}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{17}$$

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$$\mathrm{AP} = \sum_{n} (R_n - R_{n-1}) P_n \tag{18}$$

$$\mathrm{mAP} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{AP}_c \tag{19}$$

where TP denotes the number of positive samples that are correctly detected; FP denotes the number of negative samples that are detected as positive samples; FN denotes the number of positive samples that are not detected; R_n and R_{n-1} are the nth and $(n-1)th$ recall values, respectively; P_n is the precision value corresponding to the nth point; C is the total number of categories; and AP_c is the average precision achieved for the cth category.

C. Implementation Details

All the experiments below are carried out on a server with an NVIDIA RTX 4090 GPU, 24 GB of memory, and Python code running on Ubuntu 24.04. With CUDA 11.6, PyTorch 1.12, and other widely used deep learning and image processing libraries installed, our development environment is PyCharm 2022. Images from the VEDAI dataset were randomly divided into training, validation, and testing sets at a 7:1.5:1.5 ratio. We select 6000, 1000, and 868 image pairs from the DroneVehicle dataset as the training, validation, and test sets, respectively.

We train the network via the stochastic gradient descent (SGD) optimizer with some hyperparameters set as follows. The initial learning rate is 0.01, the weight decay rate is 0.0005, the optimization momentum is 0.937, the number of iterations is 300, and the number of early stopping patience rounds is set to 100, i.e., the number of rounds to be waited for after the effect is no longer boosted.

D. Ablation Study

To evaluate the effectiveness of our proposed method, we conduct a series of ablation experiments using the VEDAI dataset. In our experiment, we compared the performance of different models across various metrics and highlighted the best results by bolding the corresponding values.

1) Comparisons of Different Fusion Methods: To verify the effectiveness of the proposed fusion method, we compared two approaches: feature concatenation (Concat) and our proposed MJRNet on five different detectors.

We also compared the intermediate fusion strategy and the late fusion strategy, with the network structures shown in Fig. 9. The comparison results of different fusion methods are presented in Table I.

The model sizes and inference capabilities of different methods are evaluated based on their parameter sizes (Params) and GFLOPs. Experimental results demonstrate that, compared with the early fusion method based on simple feature concatenation, our method shows no significant difference in parameters and GFLOPs. However, by comparing multiple baseline models, we found that our method demonstrated the best detection performance in terms of mAP50 and mAP50:95. We are pleasantly surprised to find that the Ours + MJRNet results are 10.4%, 0.4%, 1.8%, and 1.7% higher than the Ours + Concat results for precision, recall, mAP50, and mAP50:95, respectively. This is a breakthrough achievement. Moreover, we observed that, compared to the two early fusion strategies, the intermediate fusion and late fusion methods exhibited significant gaps in both detection accuracy and detection efficiency. These findings indicate that, in multimodal remote sensing small object detection tasks, early fusion strategies can better preserve low-level features such as edges and textures. Intermediate fusion strategies tend to generate more redundant information, which can overshadow the features of small objects. Late fusion strategies fail to extract complementary information between modalities while introducing additional computational overhead. The above results demonstrate that the proposed MJRNet fusion strategy achieves a favorable balance between detection accuracy and computational cost.

TABLE I

COMPARISON RESULTS OF DIFFERENT FUSION METHODS ON THE VEDAI DATASET. BOLD REPRESENTS THE BEST RESULT, RED REPRESENTS THE SECOND BEST RESULT, AND BLUE REPRESENTS THE THIRD BEST RESULT

Method		Params	GFLOPs	Pre	Rec	mAP_{50}	$\mathrm{mAP}_{50:95}$
Early Fusion	YOLOv5+Concat	97.21M	247.0	74.8	63.9	73.4	46.3
	YOLOv5+MJRNet	97.43M	247.1	80.7	60.5	73.0	47.1
	YOLOv6+Concat	172.99M	611.2	60.6	47.5	53.8	33.4
	YOLOv6+MJRNet	173.05M	611.3	63.5	52.5	57.7	37.0
	YOLOv8+Concat	68.16M	258.2	73.7	64.4	72.4	46.6
	YOLOv8+MJRNet	68.22M	258.3	72.3	64.5	74.8	47.4
	YOLOv10+Concat	31.66M	171.2	76.5	64.3	72.6	47.4
	YOLOv10+MJRNet	31.72M	171.3	80.6	64.1	74.5	48.8
	Ours+Concat	45.29M	227.2	71.7	64.9	74.7	48.1
	Ours+MJRNet	45.35M	227.3	**82.1**	**65.3**	**77.9**	**49.8**
Intermediate Fusion		72.28M	362.5	53.3	51.0	53.7	33.4
Late Fusion		78.69M	395.5	70.5	63.9	71.0	44.7

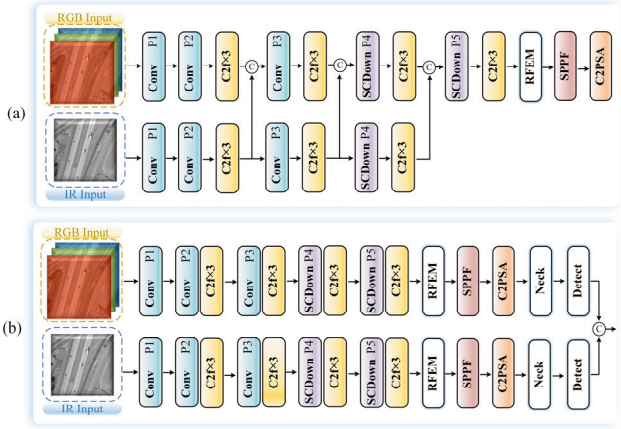


Fig. 9. Backbone networks with different fusion methods. (a) Intermediate fusion. (b) Late fusion.

2) Effectiveness of Each Component: This ablation experiment evaluates the impacts of components such as the RFEM, C2PSA, and MSIA on the performance of the object detection model. The ablation experiments conducted for different components are shown in Table II. Compared with the bare baseline, adding all three modules results in good performance.

As shown in Table II, baseline + RFEM has a 1.3% higher mAP50 than that of the original baseline, baseline + C2PSA

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TABLE II
ABLATION EXPERIMENT RESULTS ABOUT RFEM AND C2PSA, AND MSIA ON THE VEDAI DATASET. BOLD REPRESENTS THE BEST RESULT, RED REPRESENTS THE SECOND BEST RESULT, AND BLUE REPRESENTS THE THIRD BEST RESULT

#	Baseline	RFEM	C2PSA	MSIA	Params	GFLOPs	Pre	Rec	mAP_{50}	$mAP_{50:95}$
1	√	-	-	-	42.61M	225.5	73.2	61.1	70.8	45.5
2	√	√	-	-	43.28M	226.0	77.5	61.5	72.1	47.8
3	√	-	√	-	44.16M	226.7	74.8	63.0	72.7	46.3
4	√	-	-	√	43.14M	225.6	73.5	64.5	71.7	46.0
5	√	√	√	-	44.82M	227.2	77.1	63.7	73.5	47.6
6	√	-	√	√	44.69M	226.8	79.3	64.5	74.9	48.7
7	√	√	-	√	43.80M	226.1	79.5	64.6	74.9	49.5
8	√	√	√	√	45.35M	227.3	**82.1**	**65.3**	**77.9**	**49.8**

achieves 1.9% higher mAP50 than that of the original baseline, and baseline + MSIA achieves 0.9% higher mAP50 than that of the original baseline. RFEM can dynamically adjust the receptive field, enabling more effective acquisition of the background information required for each target. C2PSA further optimizes the feature map processing within the backbone network. MSIA efficiently fuses multiscale features through an iterative aggregation approach. When we add RFEM and C2PSA to the backbone network while MSIA is used to connect the backbone and the neck, we achieve excellent results in terms of both evaluation metrics (70.8% versus 77.9%, and 45.5% versus 49.8%).

3) Impact of Removing PANet: As mentioned above, although PANet enhances feature fusion through a bottom-up path, low-level features tend to be overshadowed by high-level features when the latter dominate. To investigate the impact of the PANet structure, we conducted experiments on five variants of our proposed method (MROD-YOLO): MROD-YOLOn, MROD-YOLOs, MROD-YOLOm, MROD-YOLOl, and MROD-YOLOx. As shown in Table III, we compared the metrics before and after removing PANet. We observe a noticeable improvement in the detection performance of MROD-YOLOn (53.2%→59.7% in mAP_{50}, 31.8%→37.4% in $mAP_{50:95}$), YOLOs (65.9%→67.7% in mAP_{50}, 41.0%→42.9% in $mAP_{50:95}$), MROD-YOLOm (69.1%→71.5% in mAP_{50}, 44.0%→45.6% in $mAP_{50:95}$), MROD-YOLOl (71.7%→75.0% in mAP_{50}, 45.4%→48.5% in $mAP_{50:95}$), and MROD-YOLOx (73.2%→77.9% in mAP_{50}, 48.5%→49.8% in $mAP_{50:95}$). In this study, we replaced PANet with FPN, effectively preserving the detailed information in shallow features. The additional paths and complex fusion in PANet are less direct and efficient than FPN when handling small scale targets, often introducing unnecessary noise and information interference.

Meanwhile, we observed that replacing PANet with FPN reduced the inference computational cost in MROD-YOLOn (2.46M→2.06M in Params, 9.4→8.7 in GFLOPs), MROD-YOLOs (8.40M→7.22M in Params, 27.2→24.9 in GFLOPs), MROD-YOLOm (19.41M→16.64M in Params, 71.9→68.9 in GFLOPs), MROD-YOLOl (34.92M→29.06M in Params, 149.9→146.2 in GFLOPs), and MROD-YOLOx (54.50M→45.35M in Params, 233.3→227.3 in GFLOPs). PANet, due to the introduction of additional bottom-up information transmission and path aggregation modules, leads to an increase in computational and parameter overhead. Consequently, it is less efficient than FPN in resource-constrained scenarios.

E. Comparisons With Previous Methods

We compared the performance of state-of-the-art methods, including YOLOv5 [41], YOLOv6 [43], YOLOv8 [23], YOLOv10 [15], CFT [38], DEYOLO [44], SuperYOLO [36], and our proposed MROD-YOLO method on the VEDAI dataset across various scenarios. The visualization results are shown in Fig. 10, where it can be observed that in scenarios with complex backgrounds or occlusions, other methods exhibit a significant number of missed detections and false detections. Due to the small target sizes in remote sensing images and the high similarity between different vehicle targets, existing algorithms often struggle to accurately classify the targets after localization. This issue is particularly pronounced among cars, pickups, and trucks, leading to a significant number of false detections. Our method effectively addresses this challenge and performs well in such tasks.

Table IV summarizes the performance of the above methods. The AP scores achieved by multimodal models are notably higher compared to those obtained by single models (RGB or IR) in most classes. This demonstrates that our proposed MJRNet effectively leverages early fusion strategies. Most importantly, with improvements in mAP_{50} and $mAP_{50:95}$, the increase in Params and GFLOPs is only marginal. This indicates that our method achieves a favorable trade-off between accuracy and speed.

Visualizations of some of the failure detection results are depicted in Fig. 11. An examination of the failure cases reveals that the exterior designs of cars and pickup trucks exhibit significant similarities in terms of shape, size, and color distribution, which frequently leads to confusion during detection. This often results in misclassification or ambiguous detection outcomes, particularly when the vehicles are partially occluded or captured from non-standard angles. In certain complex backgrounds, such as dense vegetation or urban structures with cluttered textures, the MROD-YOLO model may occasionally fail to detect certain targets due to the interference of environmental noise. As illustrated in Fig. 11(b), the resemblance between the vehicle and the surrounding buildings in terms of color and structural patterns caused the model to overlook the target entirely. Additionally, in scenarios with low lighting conditions or extreme weather (e.g., heavy rain or fog), the model's performance may further degrade.

F. Generalization Experiment on Dataset DroneVehicle

To validate the generalization ability of the proposed MROD-YOLO, this section compares the detection accuracy of MROD-YOLO with other methods on the public dataset *DroneVehicle*. The DroneVehicle dataset is a large-scale RGB-IR (visible-infrared) multimodal dataset for vehicle detection in drones imagery. The dataset contains a variety of shooting conditions and scenes covering different viewing angles, heights, and lighting conditions, totaling 20 000 images. The object scale in the image varies greatly, and in some scenes, the object density is high, resulting in occlusion between objects. In the visible modality, many

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TABLE III

INFLUENCE OF REMOVING THE PANET MODULE IN THE NETWORK ON THE VEDAI DATASET. BOLD REPRESENTS THE BEST RESULT, RED REPRESENTS THE SECOND BEST RESULT, AND BLUE REPRESENTS THE THIRD BEST RESULT

Method		Params	GFLOPs	Pre	Rec	mAP_{50}	$mAP_{50:95}$
MROD-YOLOn	PANet	2.46M	9.4	56.9	51.1	53.2	31.8
	Remove PANet	2.06M	8.7	64.1	56.4	59.7	37.4
MROD-YOLOs	PANet	8.40M	27.2	64.5	57.6	65.9	41.0
	Remove PANet	7.22M	24.9	71.8	58.0	67.7	42.9
MROD-YOLOm	PANet	19.41M	71.9	81.6	59.4	69.1	44.0
	Remove PANet	16.64M	68.9	81.1	58.8	71.5	45.6
MROD-YOLOl	PANet	34.92M	149.9	80.4	60.6	71.7	45.4
	Remove PANet	29.06M	146.2	80.1	64.7	75.0	48.5
MROD-YOLOx	PANet	54.50M	233.3	73.8	63.7	73.2	48.5
	Remove PANet	45.35M	227.3	**82.1**	**65.3**	**77.9**	**49.8**

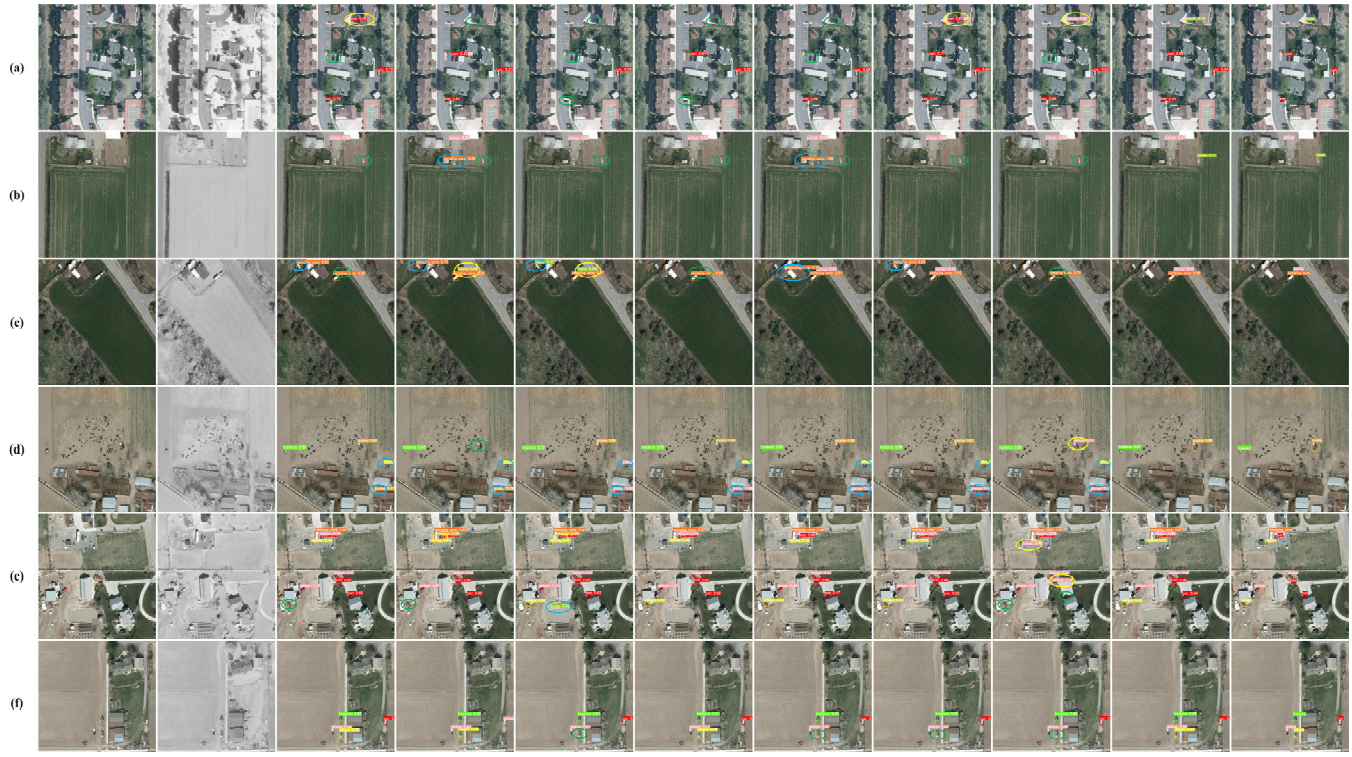


Fig. 10. Visual comparison among different detection approaches involving YOLOv5x, YOLOv6x, YOLOv8x, YOLOv10x, CFT, DEYOLO, SuperYOLO, and the proposed MROD-YOLO. The green cycles represent the missed detection, the yellow cycles represent the false detection, and the blue cycles represent the false alarm. (a)–(f) Different images in the VEDAI dataset.

nighttime images result in modality loss. In contrast, although the target boundaries in infrared images are blurred, their all-day image acquisition capability ensures that the overall shape of the targets remains relatively intact. We select 4200, 1000, and 900 image pairs from this dataset as the training, validation, and test sets, respectively. Five vehicle types, cars, trucks, buses, vans, and freight-cars, are considered.

Table V shows the comparison results produced by the different algorithms in detail, and Fig. 12 provides some visualization results. The following can be concluded.

1) The mAP_{50} and $mAP_{50:95}$ attained by the proposed MROD-YOLO model on the public DroneVehicle dataset are 70.4% and 52.7%. Our algorithm also achieves excellent detection accuracy in the generalization experiment conducted on aerial drone data relative to that of the other methods.
2) The visualization results distinctly demonstrate that the proposed MROD-YOLO method excels in reducing both missed and false detections, while also outperforming other approaches in terms of regression localization accuracy and categorical confidence, highlighting its strong overall performance.
3) We have also added comparative experiments with the latest multimodal object detection methods, CFT, DEYOLO, and SuperYOLO. It can be noted that, while our mAP_{50} is slightly lower than that of DEYOLO, our $mAP_{50:95}$ outperforms many other methods, achieving the highest overall performance. Additionally, we attained the best results for trucks and vans.
4) In summary, the proposed method demonstrates high detection accuracy on the DroneVehicle UAV aerial photography dataset, showcasing its effectiveness and strong generalizability.

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TABLE IV

PREDICTIVE PERFORMANCE COMPARISON BETWEEN OURS AND OTHER STATE-OF-THE-ART METHODS ON THE VEDAI DATASET. BOLD REPRESENTS THE BEST RESULT, RED REPRESENTS THE SECOND BEST RESULT, BLUE REPRESENTS THE THIRD BEST RESULT, AND "-" INDICATES THAT THE RESULT IS NOT REPORTED OR NOT PUBLICLY AVAILABLE

Method		Car	Pickup	Camping	Truck	Other	Tractor	Boat	Van	Pre	Rec	mAP_{50}	$mAP_{50:95}$	Params	GFLOPs
YOLOv5x	IR	89.7	80.6	72.2	80.5	44.8	64.0	52.7	75.4	71.9	64.4	70.0	44.8	97.21M	246.9
	RGB	92.3	84.4	72.4	75.1	53.9	73.5	59.4	70.1	73.0	64.8	72.6	46.9	97.21M	246.9
	Multi	93.2	84.8	69.9	75.8	63.3	76.8	51.3	68.5	80.7	60.5	73.0	47.1	97.43M	247.1
YOLOv6x	IR	88.8	70.9	60.4	59.3	24.3	61.0	20.9	52.0	67.9	48.2	54.7	34.9	172.99M	611.2
	RGB	89.8	69.9	50.1	47.3	45.8	57.8	25.9	59.8	55.6	56.0	55.8	36.2	172.99M	611.2
	Multi	93.0	75.9	62.0	61.9	39.0	54.2	26.7	48.6	63.5	52.5	57.7	37.0	173.05M	611.3
YOLOv8x	IR	88.5	79.6	74.8	**81.5**	37.9	67.4	61.5	68.7	72.4	61.6	70.0	44.8	68.16M	258.1
	RGB	93.4	84.4	73.2	70.2	55.5	**83.3**	50.3	69.4	70.9	68.6	72.5	46.3	68.16M	258.1
	Multi	91.3	85.8	68.6	77.4	59.1	72.2	61.0	**78.5**	76.0	66.6	74.2	47.4	68.22M	258.3
YOLOv10x	IR	90.7	79.7	74.1	72.1	33.3	61.9	51.5	63.2	69.6	58.4	65.8	42.8	31.66M	171.1
	RGB	92.9	84.8	73.6	73.7	63.8	68.3	51.7	66.1	74.9	64.1	71.9	47.2	31.66M	171.1
	Multi	91.7	81.0	67.9	77.4	63.6	76.9	59.3	71.4	76.6	65.1	73.7	49.7	31.72M	171.3
CFT	Multi	93.6	83.3	73.9	74.8	51.0	75.9	64.7	72.1	76.7	65.2	73.7	44.2	206.24M	-
DEYOLO	Multi	93.3	81.7	74.9	77.6	68.3	73.9	50.9	68.6	81.5	63.2	73.7	47.7	78.33M	-
SuperYOLO	IR	82.2	78.2	78.2	55.3	24.4	45.1	31.2	77.1	65.5	55.9	59.0	37.1	7.71M	-
	RGB	81.6	80.2	**83.6**	64.0	47.9	56.7	22.1	69.8	67.6	58.6	63.2	39.7	7.71M	-
	Multi	90.8	81.3	72.8	77.9	45.6	75.7	46.9	60.7	73.3	64.8	69.0	42.9	7.73M	56.2
MROD-YOLO (Ours)	IR	89.6	80.2	75.3	75.0	44.6	68.9	59.4	**78.5**	73.0	65.1	71.4	47.3	45.29M	226.5
	RGB	93.2	81.6	70.3	69.5	61.0	72.6	66.5	70.0	75.8	64.8	73.1	46.9	45.29M	226.5
	Multi	**94.0**	**89.7**	75.0	74.8	**73.2**	72.1	**68.0**	76.2	**82.1**	**65.3**	**77.9**	**49.8**	45.35M	227.3

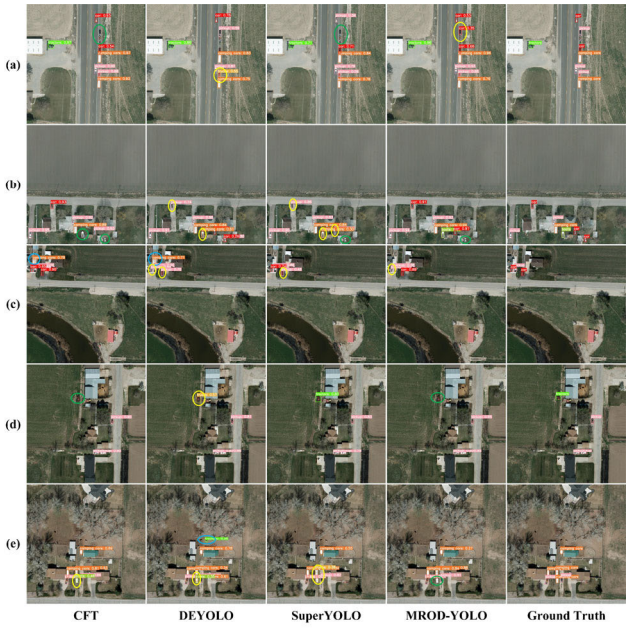


Fig. 11. Example of multimodal method detection failure. The green cycles represent the missed detection, the yellow cycles represent the false detection, and the blue cycles represent the false alarm. (a)–(e) Different images in the VEDAI dataset.

Table VI presents the comparison results of MROD-YOLO with different random seeds on the DroneVehicle dataset. The evaluation metrics include precision (Pre), recall (Rec), mAP50, and mAP50:95, which are critical for assessing the model's detection accuracy, sensitivity, and localization performance. Across five random seeds, the model demonstrates consistent performance, with precision ranging from 67.7% to 68.7%, recall from 68.6% to 70.3%, mAP_{50} from 69.9% to 70.6%, and $mAP_{50:95}$ from 50.9% to 54.4%. The mean values of these metrics are 68.3% for precision, 69.4% for recall, 70.2% for $mAP_{50\%}$, and 52.0% for $mAP_{50:95}$. These results indicate that MROD-YOLO achieves stable and reliable detection performance across different initializations, with minimal variance in the metrics, highlighting its robustness to random initialization effects.

Overall, the consistent results across random seeds validate the robustness of the proposed model in handling the DroneVehicle dataset. This stability is crucial for real-world applications where reliable and repeatable performance is essential. Future work could focus on further improving the model's localization accuracy, particularly in complex scenarios with cluttered backgrounds or small objects, to enhance its applicability in diverse environments.

V. DISCUSSION

The deployment of object detection models on resource-constrained platforms, such as embedded devices on aerial drones, poses significant challenges due to limitations in computational power, memory capacity, and energy consumption. These constraints necessitate models that are not only accurate but also efficient in terms of inference speed and memory usage. The MROD-YOLO framework, proposed in this study, aims to address these challenges by integrating multimodal data while optimizing for both accuracy and efficiency.

MROD-YOLO is designed with real-time applications in mind, particularly for scenarios where rapid detection and response are critical. The framework incorporates several key innovations that contribute to its high inference speed.

A. Lightweight MJRNet

MJRNet employs a global contextual attention mechanism to dynamically weight information from different modalities. This approach not only enhances feature extraction but also

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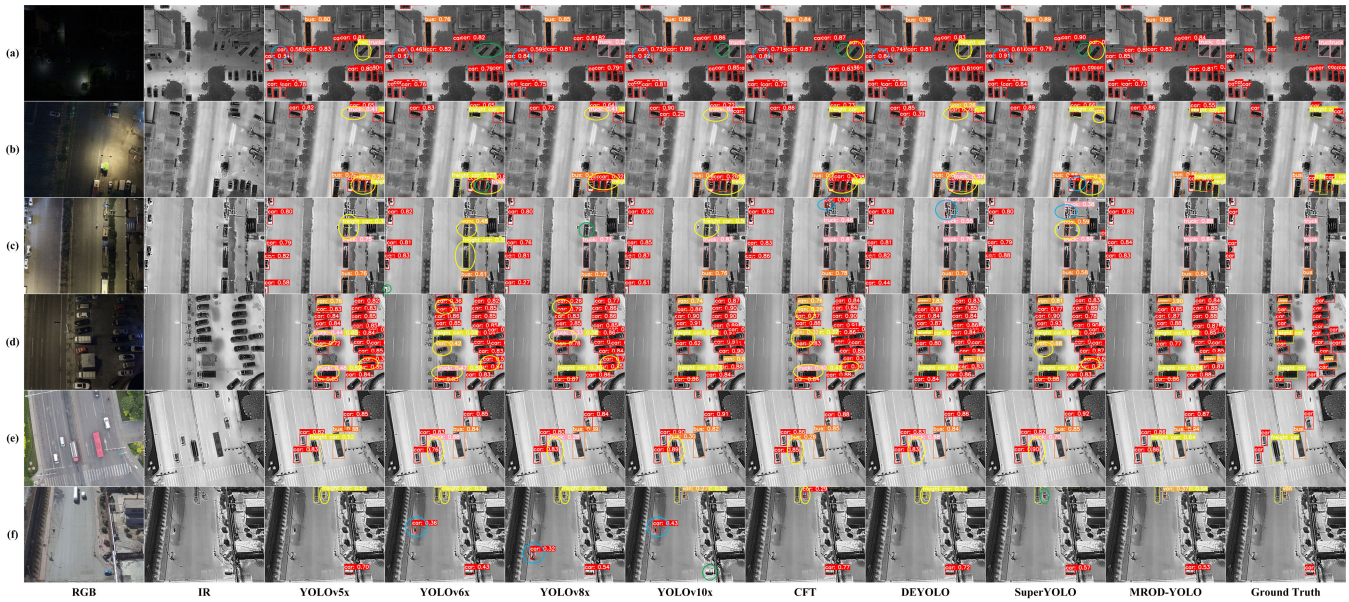


Fig. 12. Visual comparison among different detection approaches involving YOLOv5x, YOLOv6x, YOLOv8x, YOLOv10x, CFT, DEYOLO, SuperYOLO, and the proposed MROD-YOLO. The green cycles represent the missed detection, the yellow cycles represent the false detection, and the blue cycles represent the false alarm. (a)–(f) Different images in the DroneVehicle dataset.

TABLE V

PREDICTIVE PERFORMANCE COMPARISON BETWEEN OURS AND OTHER STATE-OF-THE-ART METHODS ON THE DroneVehicle DATASET. BOLD REPRESENTS THE BEST RESULT, RED REPRESENTS THE SECOND BEST RESULT, BLUE REPRESENTS THE THIRD BEST RESULT, AND "-" INDICATES THAT THE RESULT IS NOT REPORTED OR NOT PUBLICLY AVAILABLE

Method		Cars	Trucks	Buses	Vans	Freight cars	Pre	Rec	mAP_{50}	$mAP_{50:95}$	Params	GFLOPs
YOLOv5x	IR	96.8	51.3	90.1	49.5	47.7	64.9	67.9	67.1	46.4	97.20M	246.9
	RGB	87.0	48.0	82.6	47.2	38.2	60.4	59.1	60.6	36.7	97.20M	246.9
YOLOv6x	IR	96.3	41.4	85.3	33.7	46.1	60.1	60.7	60.6	42.7	172.99M	611.2
	RGB	84.0	45.5	65.1	30.4	28.4	54.3	47.3	50.9	29.7	172.99M	611.2
YOLOv8x	IR	97.0	55.1	89.8	48.8	48.7	65.6	65.9	67.9	46.9	68.15M	258.1
	RGB	86.9	48.8	80.7	45.9	42.1	61.6	57.2	60.9	37.0	68.15M	258.1
YOLOv10x	IR	96.3	48.9	88.5	43.2	49.4	68.5	61.6	65.3	46.1	31.67M	171.1
	RGB	87.4	52.1	80.8	41.9	42.8	63.7	55.4	61.0	36.9	31.67M	171.1
CFT	Multi	96.0	56.8	93.5	48.9	51.8	**76.2**	68.2	69.4	48.4	206.24M	-
DEYOLO	Multi	**97.4**	60.5	**94.7**	49.1	**52.4**	67.8	**72.0**	**70.8**	50.7	78.33M	-
SuperYOLO	Multi	96.4	50.1	89.7	47.6	50.0	70.2	66.9	66.8	45.4	7.73M	56.2
MROD-YOLO	Multi	96.7	**63.2**	93.0	**51.9**	47.3	68.7	68.6	70.4	**52.7**	45.35M	227.3

TABLE VI

COMPARISON RESULTS OF MROD-YOLO WITH DIFFERENT RANDOM SEEDS ON THE DroneVehicle DATASET

Random Seed	Pre	Rec	mAP_{50}	$mAP_{50:95}$
1	68.7	68.6	70.4	52.7
2	67.7	70.0	70.6	54.4
3	68.3	69.3	69.9	51.0
4	68.6	68.9	70.2	50.9
5	68.3	70.3	69.9	51.0
Mean Value	68.3	69.4	70.2	52.0

reduces computational overhead compared to traditional multi-branch fusion methods. By avoiding complex parallel feature extraction, MJRNet ensures that the model remains lightweight and efficient, making it suitable for platforms with limited processing power.

B. MSIA Module

MSIA efficiently fuses multiscale features through an iterative aggregation approach. By integrating high-level semantic features with low-level texture features, MSIA ensures that the model can detect small objects effectively. The iterative nature of MSIA allows for dynamic adjustment of feature interactions, further enhancing detection performance without introducing excessive computational load.

C. Replacement of PANet With FPN

The use of FPN instead of PANet simplifies the feature fusion process. FPN ensures that sufficient semantic features

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are present at shallow layers, which is crucial for small object detection. This design choice reduces the complexity of feature flow, preventing the dilution of small object features and improving overall detection efficiency.

MROD-YOLO represents a promising approach for real-time, high-precision object detection in multimodal remote sensing applications, particularly on resource-constrained platforms. Its innovative design, including the MJRNet, RFEM, C2PSA, and MSIA modules, ensures high accuracy while maintaining efficiency in inference speed and memory usage. However, further research is needed to address remaining challenges and enhance the model's scalability and adaptability. By continuing to optimize and refine the framework, MROD-YOLO has the potential to become a leading solution for real-time object detection in remote sensing imagery.

VI. Conclusion

In this article, we introduce the MROD-YOLO, an innovative and efficient multimodal object detection framework for remote sensing imagery. By integrating complementary features from optical and infrared modalities through an early fusion strategy, the proposed framework addresses the challenges of complex backgrounds, small object scales, and adverse environmental conditions. Key contributions include the development of the MJRNet and the MSIA module, which enhance feature extraction and fusion capabilities. Experiments validate that MROD-YOLO achieves state-of-the-art performance in terms of detection accuracy and computational efficiency, making it a suitable choice for real-time applications. However, further optimization is needed to reduce computational complexity while maintaining detection precision, especially for scenarios with high object density and diverse target characteristics. Future work will focus on exploring lightweight architectures and adaptive fusion techniques to enhance the model's scalability and versatility across various remote sensing tasks.

References

[1] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[2] D. Yu and S. Ji, "A new spatial-oriented object detection framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407416.

[3] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[4] J. Wang et al., "Multimodal object detection of UAV remote sensing based on joint representation optimization and specific information enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 12364–12373, 2024.

[5] Z. Zhao, J. Du, C. Li, X. Fang, Y. Xiao, and J. Tang, "Dense tiny object detection: A scene context guided approach and a unified benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, p. 1–13, 2024, doi: 10.1109/TGRS.2024.3357706.

[6] P. Li, H. Li, and G. Wang, "Towards robust detection of tiny objects in hazy remote sensing images," *Expert Syst. Appl.*, vol. 267, Apr. 2025, Art. no. 126158.

[7] Z. Zou and Z. Shi, "Random access memories: A new paradigm for target detection in high resolution aerial remote sensing images," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1100–1111, Mar. 2018.

[8] Z. Zakria, J. Deng, R. Kumar, M. S. Khokhar, J. Cai, and J. Kumar, "Multiscale and direction target detecting in remote sensing images via modified YOLO-v4," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1039–1048, 2022.

[9] H. Zheng, M. Zhang, M. Gong, A. Qin, T. Liu, and F. Jiang, "Multi-scale hierarchical feature fusion network for change detection," *Pattern Recognit.*, vol. 161, May 2025, Art. no. 111266.

[10] Q. Hou, Z. Wang, F. Tan, Y. Zhao, H. Zheng, and W. Zhang, "RISTDnet: Robust infrared small target detection network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[11] K. Wang, S. Du, C. Liu, and Z. Cao, "Interior attention-aware network for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5002013.

[12] C. Xia, S. Chen, X. Zhang, Z. Chen, and Z. Pan, "Infrared small target detection via dynamic image structure evolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5003318.

[13] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[14] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2020, pp. 213–229.

[15] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.

[16] D. Wei, Y. Liu, X. Zhu, J. Liu, and X. Zeng, "MSAF: Multimodal supervise-attention enhanced fusion for video anomaly detection," *IEEE Signal Process. Lett.*, vol. 29, pp. 2178–2182, 2022.

[17] G. Yuan, J. Song, and J. Li, "IF-USOD: Multimodal information fusion interactive feature enhancement architecture for underwater salient object detection," *Inf. Fusion*, vol. 117, May 2025, Art. no. 102806.

[18] W. Zhao, Z. Zhao, M. Xu, Y. Ding, and J. Gong, "Differential multimodal fusion algorithm for remote sensing object detection through multi-branch feature extraction," *Expert Syst. Appl.*, vol. 265, Mar. 2025, Art. no. 125826.

[19] X. Liu, C. Zhu, C. Yang, and L. Zhou, "FSFM: A feature square tower fusion module for multimodal object detection," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.

[20] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[21] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[22] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[23] H. Wang, C. Liu, Y. Cai, L. Chen, and Y. Li, "YOLOv8-QSD: An improved small object detection algorithm for autonomous vehicles based on YOLOv8," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.

[24] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.

[25] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6517–6525.

[26] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2016, pp. 21–37.

[27] Z. Li, Y. Wang, D. Xu, Y. Gao, and T. Zhao, "TBNet: A texture and boundary-aware network for small weak object detection in remote-sensing imagery," *Pattern Recognit.*, vol. 158, Feb. 2025, Art. no. 110976.

[28] H. Gong et al., "Swin-transformer-enabled YOLOv5 with attention mechanism for small object detection on satellite images," *Remote Sens.*, vol. 14, no. 12, p. 2861, Jun. 2022.

[29] Y. Lin, J. Li, S. Shen, H. Wang, and H. Zhou, "GDRS-YOLO: More efficient multiscale features fusion object detector for remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[30] W. Zhang et al., "LS-YOLO: A novel model for detecting multiscale landslides with remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4952–4965, 2024.

[31] Y. Liu et al., "YOLOv7oSAR: A lightweight high-precision ship detection model for SAR images based on the YOLOv7 algorithm," *Remote Sens.*, vol. 16, no. 5, p. 913, Mar. 2024.

[32] Y.-T. Chen, J. Shi, Z. Ye, C. Mertz, D. Ramanan, and S. Kong, "Multimodal object detection via probabilistic ensembling," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2022, pp. 139–158.

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[33] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.
[34] D. Hong, J. Chanussot, and X. X. Zhu, "An overview of multimodal remote sensing data fusion: From image to feature, from shallow to deep," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2021, pp. 1245–1248.
[35] B. Deng, P. Duan, X. Lu, Z. Wang, and X. Kang, "Hyperspectral and SAR image classification via graph convolutional fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538611.
[36] J. Zhang, J. Lei, W. Xie, Z. Fang, Y. Li, and Q. Du, "SuperYOLO: Super resolution assisted object detection in multimodal remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605415.
[37] Z. Xue and R. Marculescu, "Dynamic multimodal fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2575–2584.
[38] F. Qingyun, H. Dapeng, and W. Zhaokui, "Cross-modality fusion transformer for multispectral object detection," 2021, *arXiv:2111.00273*.
[39] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9196–9205.
[40] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.
[41] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 187–203, Jan. 2016.
[42] Y. Sun, B. Cao, P. Zhu, and Q. Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," 2020, *arXiv:2003.02437*.
[43] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.
[44] Y. Chen et al., "DEYOLO: Dual-feature-enhancement YOLO for cross-modality object detection," in *Proc. Int. Conf. Pattern Recognit. (ICPR)*, Dec. 2024, pp. 236–252.
[45] (2021). *Ultralytics/YOLOv5: V5.0*. [Online]. Available: https://github.com/ultralytics/yolov5
[46] W. Dong et al., "Fusion-mamba for cross-modality object detection," 2024, *arXiv:2404.09146*.



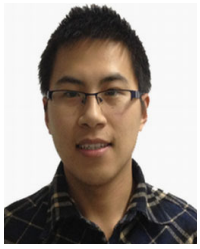
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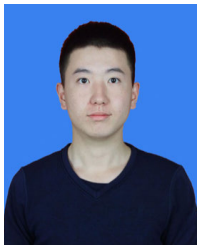
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